

Integrals – 1

Substitutions: rationalisation, Euler, the half-angle substitution

Homework

1 сентября 2026 года

Theoretical minimum. Change of variable and integration by parts; rationalisation of $R(x, x^{p/q})$; the three Euler substitutions; the half-angle substitution $t = \tan \frac{x}{2}$ together with its cheaper variants $t = \tan x$, $t = \sin x$, $t = \cos x$; hyperbolic substitutions; reduction formulas.

1. Compute $\int \sqrt{\tan x} dx$ and $\int \frac{dx}{\sin^4 x + \cos^4 x}$; in the second one show that the half-angle substitution leads to a quartic while $t = \tan x$ leads to a quadratic.
2. Decompose into partial fractions and compute $\int \frac{dx}{x^4+1}$ and $\int \frac{dx}{x^6-1}$ using roots of unity; explain how the general decomposition gives the answer for $\int \frac{dx}{x^n+1}$.
3. (Integration Bee) Speed drill:
 - a) $\int x\sqrt{2+4x} dx$ (2015);
 - b) $\int \sec x dx$ (2015);
 - c) $\int \frac{x+1}{x^2+2x+3} dx$ (2015);
 - d) $\int \frac{dx}{(x+1/x)^2}$ (2017);
 - e) $\int \frac{x^3}{1+x^2} dx$ (2013);
 - f) $\int \frac{x^5-x^3+x^2-1}{x^4-x^3+x-1} dx$ (2013).
4. (Integration Bee 2013) Compute $\int \sin x \sqrt{1+\tan^2 x} dx$, treating the sign of $\cos x$ carefully on different intervals, and explain why the naive answer is correct only on half of them.

5. Derive the reduction formulas for $I_n = \int \frac{dx}{(x^2+a^2)^n}$ and $J_n = \int \sin^n x \, dx$, and compute I_3 and J_6 .
6. (Integration Bee) Integration by parts:
 - a) $\int \sqrt{x} e^{\sqrt{x}} \, dx$ (2014);
 - b) $\int x \arctan x \, dx$ (2014);
 - c) $\int (\arcsin x)^2 \, dx$ (2014);
 - d) $\int \arctan \sqrt{x} \, dx$ (2026);
 - e) $\int \operatorname{arcosh} x \, dx$ (2017);
 - f) $\int e^{3x} \arctan(e^x) \, dx$ (2015).
7. State the rule for choosing a substitution in $\int R(\sin x, \cos x) \, dx$ according to the parity of R , and apply it to $\int \frac{\sin^3 x}{1+\cos^2 x} \, dx$, $\int \frac{dx}{1+\cos^2 x}$ (Integration Bee 2022) and $\int \frac{\cos x}{2-\cos^2 x} \, dx$ (Integration Bee 2018).
8. (Integration Bee) A quadratic under the radical: $\int \frac{dx}{\sqrt{x-x^2}}$ (2022), $\int \sqrt{12-3x^2} \, dx$ (2013), $\int \frac{\sqrt{x^2-1}}{x} \, dx$ (2014) and $\int \sqrt{x^2+25} \, dx$ (2015).
9. Compute $\int e^{ax} \cos(bx) \, dx$ in two ways: by integrating by parts twice, and through $\operatorname{Re} \int e^{(a+ib)x} \, dx$; compare the two computations.
10. (Integration Bee) A hidden derivative: notice that the integrand is $(uv)'$ or $\left(\frac{u}{v}\right)'$ and compute
 - a) $\int \frac{(\log x)(\cos x) - (\sin x)/x}{(\log x)^2} \, dx$ (2013);
 - b) $\int \frac{x+1}{x(x+\log x)} \, dx$ (2019);
 - c) $\int \frac{1+\cos x}{x+\sin x} \, dx$ (2022);
 - d) $\int \frac{1+2x^{2022}}{x+x^{2023}} \, dx$ (2023);
 - e) $\int \frac{x \log x \cos x - \sin x}{x \log^2 x} \, dx$ (2025).